

Subject Class	Registration Number	Name
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NATIONAL JUNIOR COLLEGE
SH2 PRELIMINARY EXAMINATION

BIOLOGY
Higher 1

8875/02

Paper 2 Core Paper

2 September 2009

2 hours

Additional Materials: Answer Booklet / Paper

READ THESE INSTRUCTIONS FIRST

Write your subject class, registration number and name above and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, staple your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
1	/ 15
2	/ 11
3	/ 6
4	/ 8
5 / 6	/ 20
TOTAL	/ 60

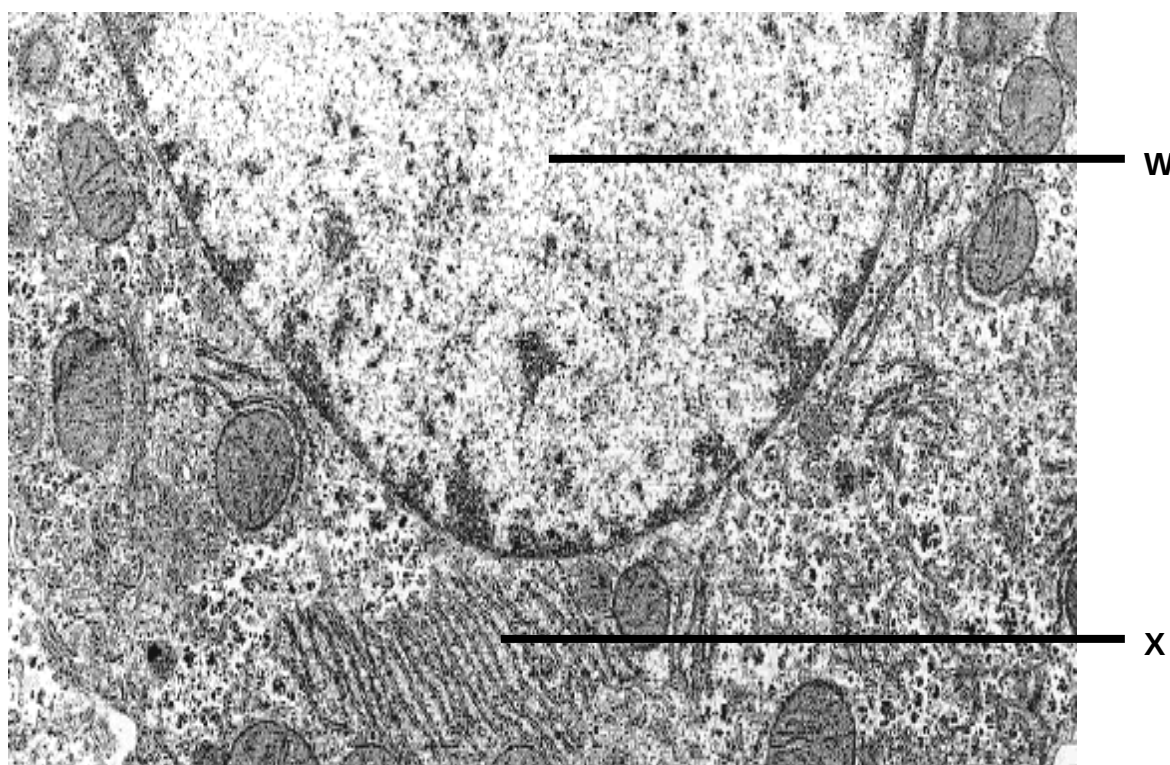
This document consists of ____ printed pages.

[Turn over

2
Section A

Answer **all** the questions in this section

1. Fig. 1.1 is an electron micrograph of a eukaryotic cell.



Picture adopted from: <http://www.ucmp.berkeley.edu/alllife/eukaryotamm.html>

Fig. 1.1

- (a) Complete the table below by listing two ways in which organelle **W** differs from that of organelle **X**. [2]

	Organelle W	Organelle X
1	<i>Double membrane bound organelle</i>	<i>Single membrane bound organelle</i>
2	<i>Absence of ribosomes studded on the membrane surface</i>	<i>Presence of ribosomes studded on the membrane surface</i>
3	<i>Presence of genetic material</i>	<i>Absence of genetic material</i>

- (b) Explain how proteins are held in the membrane of Organelle **W**. [2]

- *Hydrophobic / non-polar regions of the proteins consist of one or more stretches of non-polar amino acids*
- *Able to interact / form hydrophobic interaction with the hydrocarbon / fatty acid chains of phospholipid molecules*
- *Hydrophilic regions of the proteins (containing charged / polar R group) are exposed to the aqueous solutions on either side of the membrane*
- *Able to form hydrophilic interactions between the hydrophilic (polar) regions of the amino acids and the hydrophilic head of the phospholipid molecule*

Ehlers-Danlos syndrome is a group of inherited connective tissue disorders, which are caused by a defect in the synthesis of collagen. People with Ehlers-Danlos syndrome have soft, velvety skin that is highly elastic (stretchy) and fragile.

- (c) (i) The tripeptide sequence of glycine-glycine-alanine recurs in collagen.

The structural formula of glycine and alanine are given in Fig. 1.2.

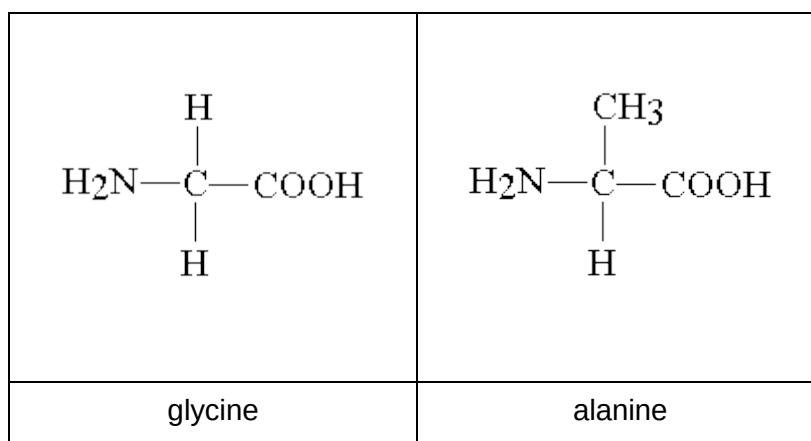


Fig. 1.2

In the space below, draw the structural formula of the tripeptide sequence. Indicate and name the bond formed between these amino acids. [2]

- *Naming of bond (peptide bond)*
- *Correct indication of bond formed*
- *Correct order*
- *Between the amino group of one amino acid and carboxyl group of another*
- *Elimination of water*

- (ii) Explain how the molecular structure of collagen contributes to its physical properties. [2]

<i>Molecular structure</i>	<i>Physical properties</i>
• <i>Each collagen molecule / tropocollagen is made up of 3 polypeptide chains</i> • <i>The 3 polypeptide chains are then wound into a right-handed coiled coil, a triple helix that is stabilized by (numerous) hydrogen bonds</i> • <i>Many of the triple helices are arranged with their ends regularly staggered</i> • <i>So that they can be cross-linked by covalent (aldehyde) bonds</i>	• <i>Organized into long parallel bundles of collagen fibers with great tensile strength</i>

- (iii) Suggest how changing one base in the nucleotide sequence of a human gene that encodes the major component of collagen would result in Ehlers-Danlos syndrome. [1]

- *Missense mutation / base pair insertion / base pair deletion / non-sense mutation*
- *Change in codon*
- *Alter the structure and function of collagen*

Fibroblasts are cells with many vital functions during development and in adult organisms, which include the synthesis of collagen. Scientists are currently researching into the effective strategies on the use of embryonic stem cells to treat Ehler-Danlos syndrome.

- (iv) Explain what is meant by stem cells. [2]

- *Unspecialized*
- *Do not perform any specialized functions*
- *Ability to self-renew*
- *Continually divide for long periods*

- *Replenish their own population and retain the complete set of genetic information*
- *Ability to differentiate into specialized cells through a process called differentiation*

(v) With reference to one example, outline the normal roles of stem cells in living organisms. [2]

- *Example: Fertilized egg (zygote)*
- *Totipotent*
- *Fertilized egg (zygote) have the ability to give rise to all the cell types that make up the body including extra-embryonic tissues*

OR

- *Example: Inner cell mass of blastocyst*
- *Pluripotent*
- *Have the ability to give rise to all of the various cell types that make up the body but not the extra-embryonic tissue*

OR

- *Example: Hematopoietic stem cells in the bone marrow (alternative answers: any appropriate adult stem cells)*
- *Ability to develop into more than one cell type of the body, giving rise to specialized cells within a particular tissue or organ*
- *Replenish worn-out / damage cells (Reject: repair cells)*
- *Able to form all the types of blood cells like red blood cells, white blood cells and platelets*

Fig. 1.2 shows two graphs showing the intracellular and extracellular concentrations of sodium (Na^+) and potassium (K^+) ions in two types of eukaryotic cells, namely frog muscle and rat liver cell.

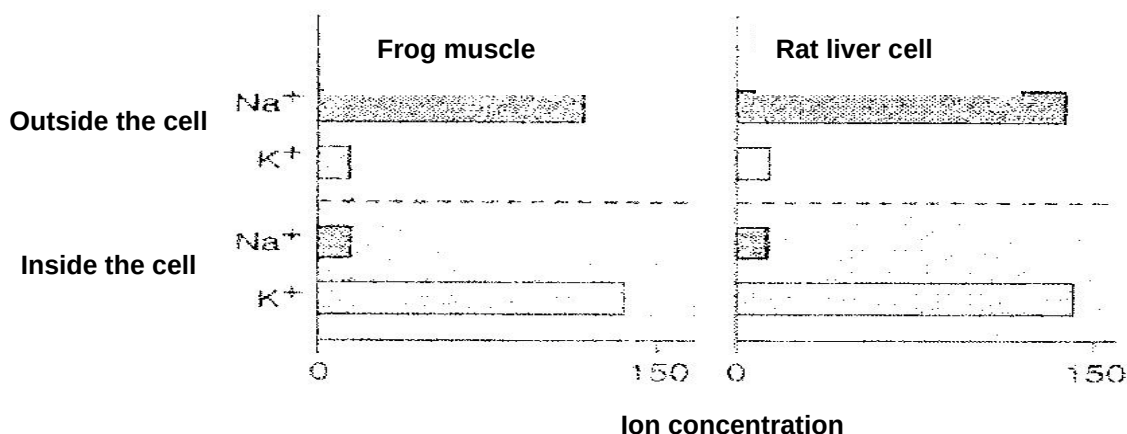


Fig. 1.2

- (d) (i) With reference to Fig. 1.2, state the general trend observed in frog muscle and rat liver cell. [1]

- *All cell types have a high Na^+ concentration outside the cell and a high K^+ concentration inside the cell*
- *Quoting of values*

- (ii) Explain how the concentration of sodium and potassium is maintained across cell membrane. [1]

- *The membrane was impermeable to ions*
- *The difference in concentration of ions are maintained with the help of specific carrier proteins which pump ions across the membrane against a concentration gradient with the expenditure of ATP*
- *One example is the sodium-potassium pump that pumps three sodium ions out of the cell for every two potassium ions pumped into the cell*

[Total: 15]

2. (a) Explain how enzymes lower the activation energy required for a reaction. [3]
- *Hexokinase / enzyme reduces the activation energy required for glucose 6-phosphate formation*
 - *When the substrate molecules / glucose and ATP collide with the active site of hexokinase in the right orientation*
 - *A enzyme-substrate complex will be formed*
 - *Contact residues of enzyme; hold the substrate molecules / glucose and ATP in the right orientation that favours the reaction*
 - *Arrangement of substrate molecules in the active site may cause physical stress to the bonds in the substrate molecules; causing them to break more easily*
 - *Catalytic residues of enzyme can change the charge on substrate molecules / alter distribution of electrons within the substrate molecules; to increase the reactivity of the substrates*
 - *Thus the initial amount of energy required for product / glucose 6-phosphate formation is lower*

Fig. 2.1 shows only part of the multi-step process of glycolysis. Phosphofructokinase-1 is one of the many enzymes that catalyse the various steps of glycolysis.

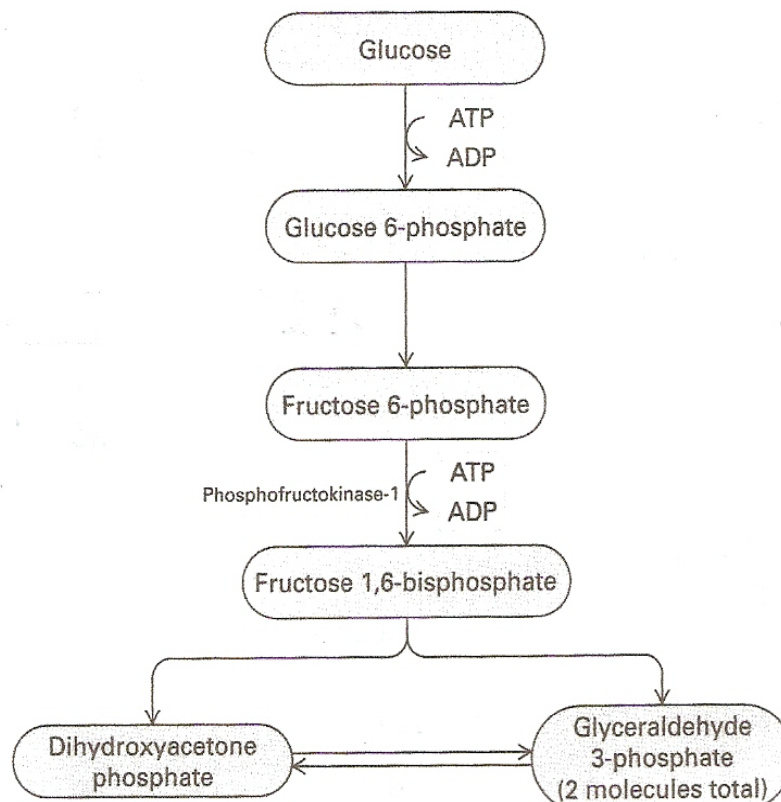


Fig. 2.1

The conversion of fructose-6-phosphate to fructose 1,6-bisphosphate is catalysed by the enzyme phosphofructokinase-1. A natural inhibitor of phosphofructokinase-1 is citrate which binds at an allosteric site on phosphofructokinase-1.

- (a) Based on the information given and from Fig. 2.1, suggest the significance of the inhibition of phosphofructokinase-1 by citrate in a respiring cell. [2]

- *Feedback inhibition*
- *Regulation of citrate level*
- *Excess citrate causes inhibition of glycolysis*
- *Less pyruvate / less acetyl coA produced*
- *Brings amount of citrate produced by Krebs cycle back to normal*
- *High level of citrate indicates Krebs cycle is at maximum rate and thus glycolysis / earlier reactions of respiration can slow down*

A different group of inhibitors has been found to act by binding to components of the electron transport chains in mitochondria and chloroplasts. These inhibitors affect both photophosphorylation and oxidative phosphorylation.

- (b) Describe two differences between photophosphorylation and oxidative phosphorylation. [2]

	<i>Photophosphorylation</i>	<i>Oxidative Phosphorylation</i>
<i>Location</i>	<i>Thylakoid membrane</i>	<i>Inner mitochondrial membrane</i>
<i>Direction of H⁺ pumping / H⁺ movement / H⁺ gradient</i>	<i>H⁺ is pumped from stroma to thylakoid space / lumen, moves back into stroma through ATP synthetase</i>	<i>H⁺ is pumped from matrix to intermembranal space, moves back into matrix through ATP synthetase</i>
<i>Electron donors</i>	<i>Water and photosystems</i>	<i>Respiratory substrates / NADH and FADH₂</i>

Rubisco (RuBP carboxylase) activity can be reduced by inhibition. Oxygen is an inhibitor of Rubisco. Fig. 2.2 shows the rate of photosynthesis of without inhibition (Graph **A**) and with inhibition by oxygen (Graph **B**).

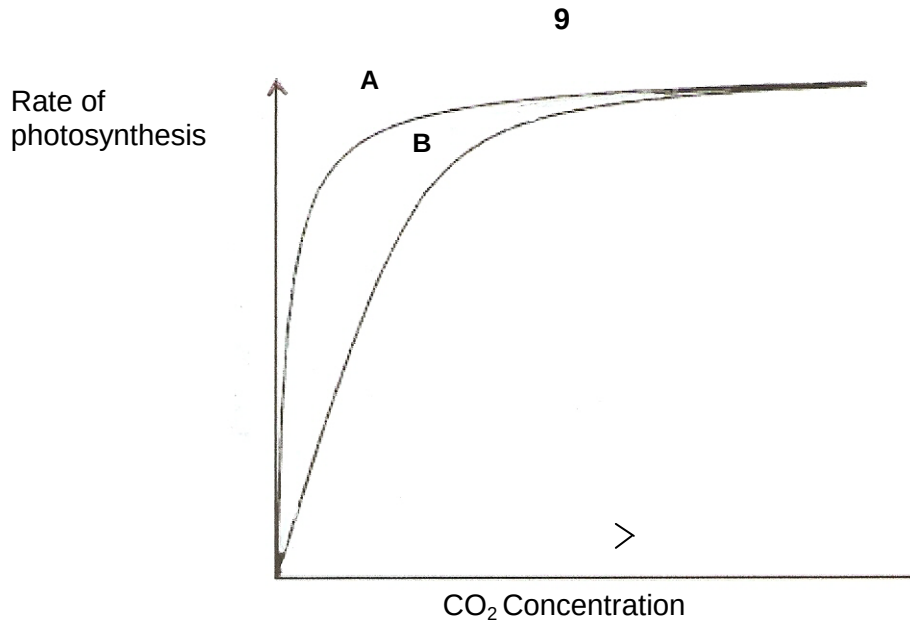


Fig. 2.2

- (c) Based on Fig 2.2, state, with reasons, whether oxygen is a competitive or non-competitive inhibitor. [2]

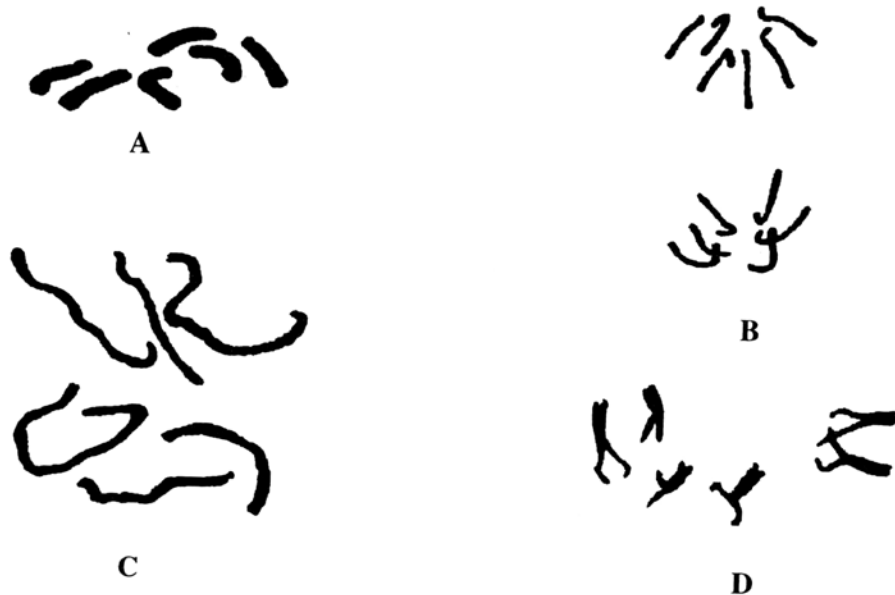
- *Competitive inhibitor*
- *Rate of photosynthesis reaches V_{max}*
- *Increasing substrate concentration (CO_2) reduces the effect of the inhibitor as at higher concentrations CO_2 will out-compete O_2 to bind to the active sites*

- (d) Account for the decrease in a plant's productivity when Rubisco has been inhibited. [2]

- *Fixation of CO_2 decreases / rate of combination of CO_2 and RuBP decreases*
- *Fewer unstable 6C compounds / GP molecules formed per unit time / ref to rate*
- *Rate of G3P production decreases*
- *Amount of G3P channeled out to form glucose decreases*
- *Thus less glucose formed from photosynthesis*

[Total: 11]

3. The microphotographs below show four stages in mitosis occur in an organism A.



- (a) Give the correct sequence of the four stages shown above.

1 C 2 A 3 D 4 B

[1]

- (b) Explain how mitosis results in daughter cells containing copies of genetic information identical to the parent cell. [2]

- *Prior to mitosis, cells undergo DNA synthesis where the original DNA molecules replicate, forming 2 identical daughter DNA molecules*
- *Each chromosome consists of two identical sister chromatids held together at the centromere*
- *During anaphase, the centromere divide and the sister chromatids separate; and are pulled to opposite poles of the spindle*
- *Daughter cells formed are genetically identical*
- *Because the chromosomes have the same base sequence (or same alleles)*
- *Since they were a product of semi-conservative DNA replication*

- (c) State the number of chromosomes in a somatic cell of organism A.

- 6

[1]

- (d) Explain how one named factor increases the chance of cancerous growth. [2]
- *Exposure to ionizing radiation (from X / gamma / nuclear radiation)*
 - *Overexposure to ionizing radiation may cause mutation of the DNA / genes that control normal rate of cell division*
 - *Gene mutation occurs during DNA replication where a base(s) is/are wrongly substituted*
 - *For example, proto-oncogenes that stimulate cell division in normal cells are mutated, becoming oncogenes*
 - *Oncogenes are more active, causing cells to divide more often*
 - *Alternatively, tumor suppressor genes that normally inhibit cell division are mutated, becoming less active in its inhibition mechanism*
 - *In both cases, normal cell cycle regulation is disrupted and cells divide uncontrollably leading to cancer and tumor formation*

[Total: 6]

- 4 Patients with anti-thrombin (AT) deficiency can be treated with human AT, which is manufactured by recombinant DNA technology. Fig. 4.1 outlines the procedures for cloning human AT cDNA in a DNA plasmid from bacterium. Suitable recipient bacteria are then treated to form bacteria with recombinant plasmids.

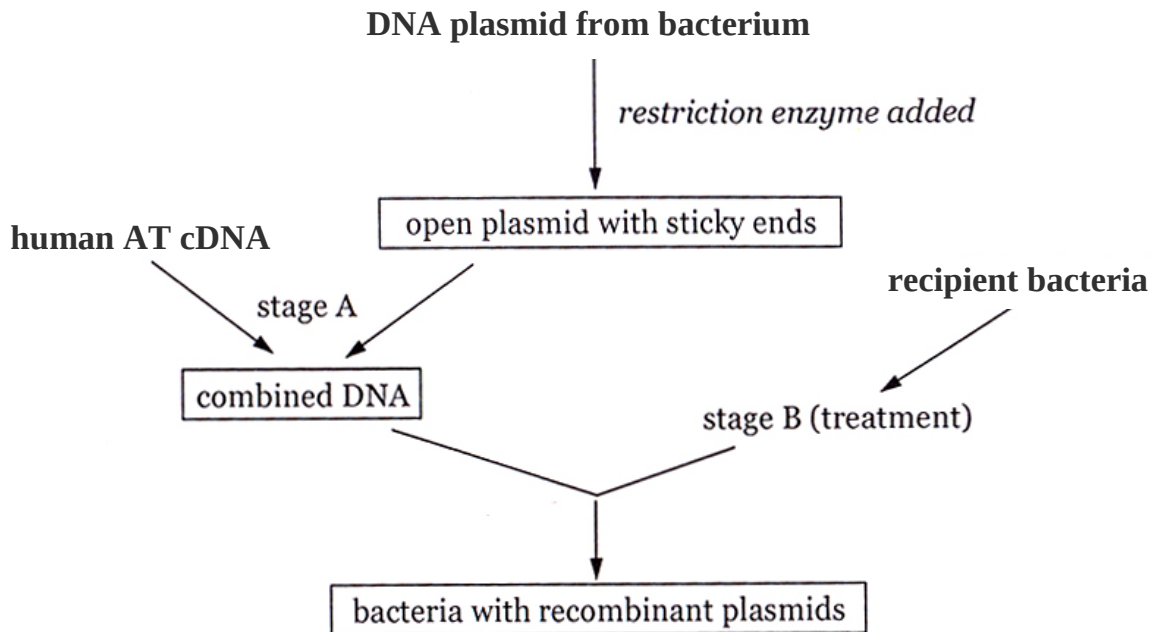


Fig. 4.1

- (a) State **two** property of the DNA plasmid that allows it to be used as a DNA cloning vector. [1]
- *one origin of replication / at least one selectable marker / at least one unique restriction site*
- (b) Describe the natural functions of restriction enzymes. [1]
- *E. coli can protect itself from infection by bacteriophages by using its restriction enzymes such as EcoRI*
 - *Fragment any foreign / viral DNA as soon as the latter enters the bacterial cell*
- (c) Name the enzyme used in stage A to form the combined DNA. [1]
- **DNA ligase**

- (d) Describe how bacteria can be screened for the presence of recombinant plasmids based on β -galactosidase activity. [3]

- *Bacterial plasmids or Expression / Cloning vectors, each of which contains 1 antibiotic resistance gene (e.g. Amp^r) and unique cloning sites within the lacZ gene are used to transform bacteria that lack R plasmids*
- *The bacteria are then plated on an agar plate containing ampicillin to eliminate those with no uptake of the cloning vectors*
- *The same agar plate also contains X-Gal and IPTG whereby IPTG induces the production of functional β -galactosidase that catalyses the hydrolysis of X-Gal to a blue product*
- *Clones of bacteria that contain the gene of interest will appear white while those that do not contain the gene of interest will appear blue*

- (e) The use of genetically modified organisms has ethical and social implications. Outline these implications. [2]

- *GM food might cause food allergies*
- *Genes from GM plants may transfer through cross-pollination*
- *May lead to a reduction in biodiversity*
- *Seen as 'tempering with nature' / religious objections*
- *Objections to consuming animal genes in plants / vice versa*
- *Development may benefit rich countries at the expense of poorer countries*
- *World food production may become dominated by small number of large companies*

[Total: 8]

Section B

Answer only **one** question from this section

5. (a) Describe, with examples, how the molecular structure of phospholipids is related to their function in the membrane. [6]
- *Comprises of a phosphate group, glycerol backbone and 2 fatty acid molecules*
 - *Ester bond formation between glycerol and each fatty acid molecule*
 - *Amphipathic*
 - *Containing a polar hydrophilic and a hydrophobic non-polar region within the same molecule*
 - *Major component of the plasma membrane by forming bilayer*
 - *Polar phosphate heads are on the outside of the bilayer, in contact with the surrounding aqueous medium*
 - *Fatty acid chains, being hydrophobic and non-polar, point towards the interior of the bilayer, away from the aqueous medium*
 - *Partially permeable – Permeable to lipid-soluble substance*
 - *Impermeable to polar ions (hydrophilic molecules)*
 - *Serve as a barrier to retain cell content / compartmentalization*
 - *Influence the fluidity of membrane*
 - *The more unsaturated fatty acid chains are, the more fluid the membrane is*
 - *Prevent close packing of the molecule and decrease the interaction between adjacent fatty acid chains*
 - *Likewise, lipids with shorter fatty acid chains are more fluid*
 - *Shorter chain length reduces the tendency of the hydrocarbon tails to interact with one another*

(b) Outline the main ways in which HIV protease differs from DNA.

[7]

	<i>HIV protease</i>	<i>DNA</i>
<i>Monomers involved</i>	<i>Amino acids</i>	<i>Deoxyribonucleotides</i>
<i>Types of monomers</i>	<i>20 different monomers</i>	<i>4 different types of monomers</i>
<i>Bonds involved</i>	<i>Monomers are joined by peptide bonds</i>	<i>Monomers are joined by phosphodiester bonds</i>
<i>Presence of complementary base pairing</i>	<i>Absence of complementary base pairing</i>	<i>Presence of complementary base pairing via hydrogen bonds. For example, A=T and G≡C</i>
<i>General structure</i>	<i>Spherical shape / homodimer. Structure held by hydrogen bonds, ionic interaction, disulphide bonds, and hydrophobic interactions</i>	<i>Helical structure Structure is held mainly by hydrogen bonds</i>
<i>Function</i>	<i>Metabolic / Catalytic: Enzyme that hydrolyses inactive viral polyproteins produced by HIV in its host cell into several active proteins that are essential for viral assembly and subsequent activity</i>	<i>DNA contains the hereditary material / genetic information for living organisms</i>
<i>Presence of active site</i>	<i>Presence of active sites / pockets for binding of substrates</i>	<i>Absence of active site</i>
<i>Presence of flaps</i>	<i>Presence of 2 flexible flaps</i>	<i>Absence of flaps</i>
<i>Arrangement</i>	<i>The 2 polypeptide chains are not anti-parallel</i>	<i>The 2 polynucleotide strands are anti-parallel</i>
<i>Similarity between the polypeptide chains / polynucleotide strands</i>	<i>The 2 polypeptide chains are identical</i>	<i>The 2 polynucleotide strands are not identical</i>

- (c) With reference to specific examples, explain the significance of having a variety of carbohydrates in eukaryotes. [7]

- *Carbohydrates are molecules containing carbon, hydrogen and oxygen only*
- *Can be classified into how many repeating units they contain (e.g. monosaccharides, disaccharides, etc)*
- *Or the number of carbon atoms they possess (i.e. triose, pentose etc)*
- *Triose sugars contain only 3 carbon atoms*
- *Most important biological example is 3-phosphoglyceraldehyde (PGAL) / 3-PG*
- *Molecule is found in the glycolysis pathway of glucose metabolism*
- *As well as in the Calvin cycle of photosynthesis*
- *Pentose sugars have 5 carbon atoms*
- *Three important pentose sugars – ribose, deoxyribose, and ribulose*
- *Ribose – backbone for the synthesis of RNA*
- *Deoxyribose – backbone for the synthesis of DNA*
- *Ribulose – used in the production of ribulose biphosphate (RuBP)*
- *Which serves as a carbon dioxide acceptor in the light-independent reaction of photosynthesis*
- *Hexose sugars contain 6 carbon atoms / also known as monosaccharides*
- *Include glucose, galactose, and fructose*
- *Glucose is the basic respiratory substrate used by living organisms*
- *Monosaccharides are linked together by forming a glycosidic bond between them*
- *Through condensation reaction*
- *To form disaccharides (linking 2 monosaccharides together)*
- *Polysaccharides formed by condensation of several hundreds of monosaccharides*
- *Large molecular weight*
- *Insoluble*
- *Starch and glycogen*
- *Formed by condensation of alpha-glucose*
- *Compact, helical or branched structures*
- *Storage molecules*
- *Cellulose*
- *Formed by the condensation of beta-glucose*
- *Cross-linking via hydrogen bonds to form strong structural fibrils*

6. (a) Describe the eukaryotic processing of pre-mRNA.

[6]

{5' cap}

{Description}

- *Addition of 5' cap*
- *After transcription / 1st of the 3 processes to take place*
- *Catalysed by a capping enzyme*
- *7-methylguanosine added / modified guanine*

{Function}

- *Facilitates the export of mRNA from nucleus to cytoplasm / through nuclear membrane*
- *Small subunit of ribosome may bind to it to initiate translation*
- *Protects RNA transcript from degradation by exonucleases*

{poly(A) tails}

{Description}

- *Endonuclease recognizes polyadenylation sequence*
- *Endonuclease cuts pre-mRNA several nucleotides beyond 3' end of polyadenylation sequence*
- *Addition of poly(A) tails to 3' ends / polyadenylation to 3' ends*
- *Catalysed by poly-A polymerase*
- *Poly(A) tails can be about 200 nucleotides long*

{Function}

- *Protects mRNA from degradation by exonuclease*
- *Facilitates the export of mRNA from nucleus to cytoplasm / through nuclear membrane*

{mRNA splicing}

{Description}

- *mRNA splicing*
- *introns / intervening sequences are spliced out by the action of a spliceosome*
- *pre-mRNA can be spliced in different ways / alternative mRNA splicing*
- *Introns can be removed separately or in combination*
- *If two successive introns are removed together, the exon between them will be removed as well / opportunity to modify the coding sequence of RNA by deleting some of its exons*

- *Alternative mRNA splicing allows a single gene to encode different polypeptides / different RNA transcripts to be produced by a single gene / protein variants from a single gene*

(b) Define control elements and explain how they influence transcription in the eukaryotic genome. [7]

(Definition)

- *Control elements are segments of non-coding DNA that help to regulate transcription by binding to transcription factors or to RNA polymerase*

(Promoters)

- *Promoters bind to RNA polymerase*
- *RNA polymerase / general transcription factors (GTFs) recognizes the TATA box within the promoter*
- *Promoter may also bind to general transcription factors (GTFs) to form a transcription initiation complex*
- *When RNA polymerase and TFs bound, allows for transcription*
- *General transcription factors (GTFs) bound to promoter or proximal control elements mediate the binding of RNA polymerase to the promoter*
- *Interaction of promoter or proximal control elements to GTFs and GTFs interaction with RNA polymerase usually leads to only a basal level of transcription / production of a little RNA*

(Enhancers & Silencers – in general)

- *Enhancers and silencers bind to specific transcription factors*

EITHER

- *Enhancers bind to activators to upregulate transcription / increase transcription*
- *Silencers bind to repressors to inhibit or downregulate transcription / stop transcription*

OR

- *Enhancers or silencers bound to specific transcription factors (STFs) allow for the regulation of the level of gene expression*

(Distal control elements – Enhancers)

- *Distal control elements / some enhancers and silencers are located a long distance away from the promoter / thousands of nucleotides upstream or downstream of the promoter*
- *When enhancer (distal) binds with an activator, it causes the binding of a DNA-bending protein*
- *This brings the bound activator / STF closer to the promoter / Enables the enhancer to influence a promoter hundreds or even thousands of nucleotides away*
- *Activators then bind to certain GTFs and mediator proteins / these multiple protein-protein interactions help assemble and position the transcription initiation complex on the promoter / allow RNA polymerase to bind more efficiently or with higher affinity*
- *Enhancers binding to activators may cause histone acetylation / increase gene accessibility to RNA polymerase and TFs / decondense DNA*
- *Increasing the rate of transcription*

(Distal control elements – Silencers)

- Silencers bound to repressors can also inhibit RNA polymerase binding
- Silencers bound to repressors may cause histone deacetylation / decrease gene accessibility to RNA polymerase and TFs / condensing DNA
- Decreasing the rate of transcription

(Combinatorial control)

- Combinatorial control of genes occurs because each gene has a unique combination of CE
- Depending on the different types of TF / activators found in each cell, genes will be switched on
- Determines cell-type specific transcription

(Coordinate control)

- Coordinate control of genes with related function occurs because the group of genes share the unique combination of CE

- (c) Describe how the differences in the structure and organization of prokaryotic and eukaryotic genomes affect their control of gene expression. [7]

		<i>Prokaryotic Genome</i>	<i>Eukaryotic Genome</i>
<i>Structure</i>	<i>Level of condensation</i>	• DNA not highly condensed / Not bound to histones • Because genome size is smaller	• Many levels of condensation of DNA / DNA bound to histones • Because genome size is larger
	<i>Significance</i>	• No need to decondense chromosome / uncoil DNA from histones before transcription can occur	• Chromatin must be decondensed / have a looser conformation for transcription to occur so that RNA polymerase & transcription factors can bind to DNA
		• Changes in chromosome structure not used as method to regulate transcription	• Rate of transcription can be regulated by histone acetylation • resulting in conversion between euchromatin and heterochromatin, affecting the ease of transcription
<i>Structure / Organisation</i>	<i>Chromosomes / Genomes bound by membrane</i>	• Not bound by membrane / genome in nucleoid	• Bound by the nuclear membrane / genome in nucleus
	<i>Significance</i>	• Transcription and translation occur simultaneously	• Transcription occurs before translation / transcription occurs within the nucleus and translation outside the nucleus / transcription and translation occur in different locations in the cell

		• <i>Post-transcriptional control of gene expression not possible</i> • <i>Thus mRNA is more easily degraded / less stable</i>	• <i>Post transcriptional control of gene expression is possible</i> • <i>Eg. Alternative splicing / mRNA modifications before translation – 5' cap or poly-A tail</i> • <i>Thus mRNA is less easily degraded / more stable</i>
Structure / Organisation	Presence / Absence of introns	• <i>No introns</i>	• <i>Presence of introns</i>
	Need for mRNA splicing & allows for alternative splicing	• <i>No splicing of mRNA is needed</i> • <i>No alternative splicing possible</i> • <i>Only one mRNA / protein product per gene</i>	• <i>Splicing of pre-mRNA is needed / exons / coding regions must be spliced together</i> • <i>Allows for alternative splicing giving rise to varying mature mRNA molecules and hence multiple possible protein variants from the same gene</i>
Organisation	Presence / absence of operon	• <i>Genes coding for enzymes with related function grouped in an operon</i>	• <i>Dispersed genes coding for enzymes with related function each controlled by an individual / separate promoter</i>
	Need for coordinate control, eukary genes having shared control elements	• <i>Produces polycistronic mRNA</i> • <i>Allows coordinated control of metabolic enzymes involved in the same pathway</i>	• <i>Produces monocistronic mRNA</i> • <i>Coordinated control occurs via binding of transcription factors to control elements shared by genes with related functions (when genes are on diff chrom)</i> • <i>Coordinated control also occurs by DNA methylation / histone acetylation that causes all the genes in the same region to be made available or unavailable for transcription (when genes w related fns are close together on the same chrom)</i>

		• <i>All gene products of an operon are produced in the same amount</i>	• <i>Each gene product can be produced in different amounts or not at all</i>
Organisation	<i>Presence / Absence of (distal) control elements</i>	• <i>No distal control elements or its regulatory sequences are close to promoter</i>	• <i>Presence of distal control elements</i>
	<i>Allows for more fine-tuned control of transcription</i>	• <i>Simple control possible, either transcription switched on or off</i> • <i>Level of expression cannot be finely tuned / regulated as specifically</i>	• <i>Allows for fine tuned control of gene expression / allows for different levels of expression of a particular gene depending on the activators and repressors present in the cell at a specific time</i>

[Total: 20]

